



One Health: Soil-Plant-Animal-Human Continuum

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Abstract

Soil underpins the One Health continuum, connecting plant, animal, and human well-being. In India, soil contamination, nutrient imbalance and chemical overuse are affecting crop productivity, livestock health and food safety. This article examines how soil disturbances propagate through plants and animals to impact human health, highlighting risks from heavy metals, pesticide residues, and antimicrobial resistance. It emphasizes the role of integrated monitoring tools and sustainable soil management practices. Adopting a One Health approach, the article advocates cross-sector collaboration to safeguard soil health, ensure safe food production and strengthen resilient farming systems.

Keywords: One Health, Soil health, Agrochemicals, Heavy metals, Pesticide residues and Antimicrobial resistance

Introduction:

The One Health concept recognizes that human, animal, plant and environmental health are interdependent. In India's agrarian landscape, soil quality is a keystone: healthy soils support nutritious crops and livestock, which in turn sustain human health. Conversely, soil contamination or degradation can propagate through the food chain: chemicals or pathogens in soil may be taken up by plants, accumulate in animals, and ultimately affect people. India already faces serious soil health challenges: most soils are low in nutrients (average organic carbon ~0.54%) and roughly 30% of land is degraded. Excessive erosion - about 3 billion tonnes of fertile soil lost annually - and chemical imbalances (e.g. overuse of NPK without micronutrients) further weaken soils. Inappropriate agrochemical use not only depletes soil fertility, it "poses significant risks to human health and ecosystems". Against this backdrop, India's farmers and authorities are increasingly recognizing the soil-plant-animal-human continuum. For instance, the FAO and Government of India promote soil health cards and sustainable fertilizer practices to maintain fertility and food quality. This review examines evidence - from Indian case studies, ICAR research and national/international reports - showing how soil issues impact crops, livestock and public health, highlighting One Health challenges in the Indian context.

Soil degradation and nutrient imbalance: India’s soils face both physical erosion and chemical degradation. An estimated 3 billion tonnes of topsoil are lost each year by wind and water erosion, stripping fields of organic matter and nutrients. Many soils are inherently nutrient-poor (e.g. low organic C ~0.54%), and decades of single-crop cultivation with high N-P-K fertilization (Green Revolution practices) have created imbalances. Studies note that “overuse, misuse, or imbalanced application of fertilizer nutrients” has led to nutrient accumulation and leaching, causing environmental pollution. Such practices intensify soil and water pollution (eutrophication) and threaten biodiversity and health. For example, excess urea application in Punjab has elevated nitrate levels in soil and waterways. Nutrient-poor soils also yield micronutrient-deficient crops, exacerbating hidden hunger: FAO reports that multi-nutrient deficiencies in soil adversely affect human nutrition in local diets. In summary, soil degradation reduces crop productivity and nutritional value, undermining plant and human health before plants even enter the continuum.

Heavy metal contamination (Soil-Plant-Animal): Industrial and agricultural pollutants are elevating toxic metals in many Indian soils. Rajaganapathy *et al.* (2011) reported that heavy metals like Cd, Pb and Hg are “a serious health concern” in India, with “high levels... reported in different parts of India”. These metals persist in soil and enter crops; for instance, an environmental study in Punjab’s Ropar wetland found all soil samples with high cadmium and cobalt, and rice and mustard crops accumulating Cr, Co, Pb and Cd. Bio concentration factors showed rice hyper-accumulating Cr and Cu and maize accumulating Cu. Health risk assessments indicated that consuming these crops could pose cancer and non-cancer risks for children (hazard indices >1 for Cr, Co, Pb). Such examples illustrate the soil-plant link: contaminated soils produce contaminated food. The problem extends to animals: livestock fed on polluted fodder or water suffer toxicity. Heavy metal exposure in animals accumulates in organs and milk, posing risks up the chain. Thus, soil contamination with metals breaks the continuum: plants accumulate toxins, herbivores fall ill and humans eating animal products or crops face serious health hazards.

Pesticide residues and agrochemicals: The persistent use of pesticides has left a legacy in Indian soils and crops. According to FAO, “inappropriate use of pesticides and fertilizers not only threatens soil and water quality but also poses significant risks to human health and ecosystems”. Monitoring data confirm widespread residues: India’s national survey (2006-2012) found pesticide traces in ~9.2% of food and environmental samples, with 1.5% above safety limits. More recently, FSSAI reported that 86,401 food samples (2022–25) were tested and ~2.8% exceeded permissible pesticide limits. Vegetables and fruits often carry organophosphate or organochlorine residues from soil and spraying. For example, DDT and Endosulfan remnants continue to be detected in Yamuna floodplain soils and vegetables in Haryana. These residues affect plant physiology and yield (crop stress, slowed growth) and enter the food chain. Livestock feeding on treated stubble or grazing near sprayed fields ingest pesticides, which can impair their reproduction and contaminate milk/meat. Human consequences include acute poisoning (farmers) and chronic illnesses (neurological, endocrine disruption) in consumers. In sum, agrochemical overuse has degraded soil biology and left chemical residues in crops, linking soil health to food safety.

Antimicrobial use and soil-human health: Antibiotics and growth promoters used in Indian livestock add another dimension to the continuum. A large fraction of veterinary antibiotics is excreted without metabolization; studies estimate ~75% of antibiotics fed to livestock can enter the environment via manure or runoff. Once in soil, these drugs disrupt microbial communities and select for resistant bacteria. Indeed, India has among the highest rates of antibiotic resistance in humans and animals, making AMR a critical One Health issue. Taneja & Sharma note that “injudicious use of antimicrobials and inadequate treatment of waste waters are important drivers of AMR in India”. Experiments further show that applying animal manure compost significantly increases antibiotic-resistant bacteria in soil. Thus, resistant pathogens can cycle from animals to soil to humans (e.g. through contaminated water or fresh produce). This soil-animal-human loop of AMR undermines treatment of infections in people and livestock alike. In India’s context - with dense populations of humans and livestock - the spread of soil-mediated AMR emphasizes the need for integrated policies across agriculture and health sectors.

Implications for crop, livestock and public health: Disruptions in soil health ripple through the food system. Soil nutrient depletion translates to lower crop yields and poorer nutritional quality (e.g. micronutrient-poor grains). Soil pollutants compromise crop safety, as seen in heavy-metal and pesticide accumulation in rice, wheat and vegetables, raising food-safety concerns. For livestock, contaminated soil means tainted fodder and drinking water, leading to reduced fertility, growth rates and higher disease; outbreaks of poisoning (e.g. the lead-related cattle mortalities) impose economic losses on farmers. Ultimately, humans face the combined burden: micronutrient deficiencies from degraded soils, chemical and microbial contamination from the food chain, and harder-to-treat infections from environmental AMR. FAO and WHO stress that healthy soils are essential to “anticipate, prevent, detect and control diseases that spread between animals and humans”. In India, where agriculture, livestock and human health sectors intersect, addressing one facet (e.g. soil contamination) requires coordinated action in the others.

Conclusion

The Indian evidence underscores that soil health underlies One Health. Soil contamination and mismanagement do not remain isolated: they cascade into plants, animals and people. Key challenges include heavy-metal soil pollution near industries, nutrient imbalances from fertilizer misuse, persistent pesticide residues, and the spread of antibiotic residues through manure. These issues manifest as crop yield decline, livestock illness, tainted food and rising public health risks. Mitigation demands integrated strategies. For example, researchers advocate integrated nutrient management (combining organic manures with balanced chemical fertilizers) to sustain soil fertility and reduce residues. India’s Soil Health Card scheme and ICAR initiatives aim to guide balanced fertilization. Pesticide regulations and monitoring (as in the MPRNL scheme) must continue enforcing safety. Controlling AMR calls for judicious antibiotic use in animal husbandry and proper waste treatment. In essence, maintaining soil organic matter and biodiversity, minimizing toxic inputs, and monitoring food safety are critical. A One Health approach - linking agricultural extension, environmental protection and health programs - is essential. By safeguarding soil quality and closing the loop between fields and communities, India can protect crop productivity, livestock welfare and human health simultaneously.

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